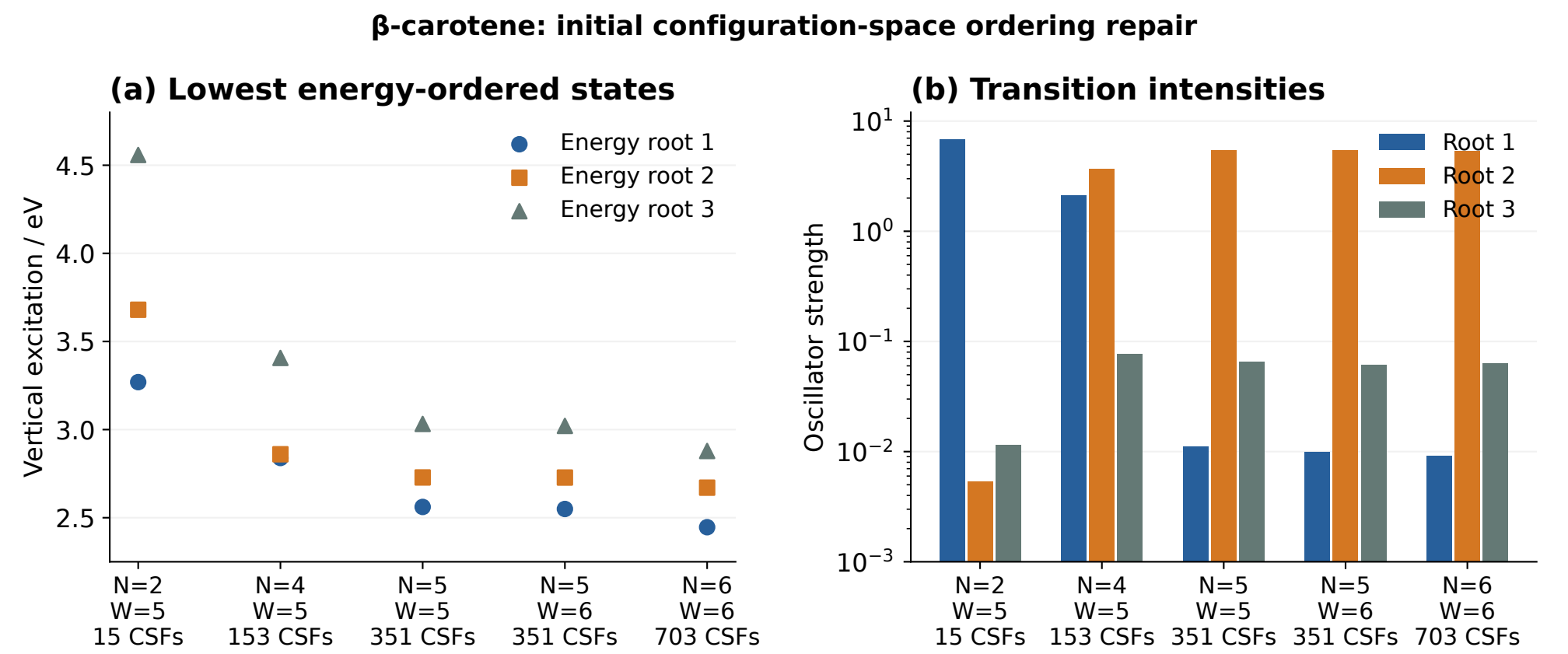
β-carotene: initial configuration-space ordering repair
(a) Lowest energy-ordered states
Vertical excitation / eV
4.5
4.0
3.5
3.0
2.5
Energy root 1
Energy root 2
Energy root 3
N=2
W=5
15 CSFs
N=4
W=5
153 CSFs
N=5
W=5
351 CSFs
N=5
W=6
351 CSFs
N=6
W=6
703 CSFs
(b) Transition intensities
Oscillator strength
$10^{1}$
$10^{0}$
$10^{-1}$
$10^{-2}$
$10^{-3}$
Root 1
Root 2
Root 3
N=2
W=5
15 CSFs
N=4
W=5
153 CSFs
N=5
W=5
351 CSFs
N=5
W=6
351 CSFs
N=6
W=6
703 CSFs